



INTERNSHIP - RTL2GDSII

PLACEMENT

in

PHYSICAL DESIGN

using

INTEGRATED CIRCUIT COMPILER (ICC2)

Only for chip to startup (C2S) Training

PLACEMENT

Placement is the process of finding an appropriate physical location for each cell in the block.

Tool determines the location of each component of design inside core area in such a way that the design should have **minimal congestions and the best timing**

• **TRIAL ROUTING – GLOBAL ROUTING CELLS- TRACKS**

Checks before Placement

- `check_design` -checks pre_placement_stage

To check the following :-

- No Pin left disconnected
- No floating nets
- No S.C/O.C connections
- Proper Pin/Port Placement
- Macros(if available) should be in fixed location (Placement for Std.Cell)
- Power routes must be free from DRCs



Goals and Inputs for Placement

GOALS

- Better Timing, Reduced Area and better Power Optimization
- Minimum Cell density and Pin Density
- Minimize Congestion and Congestion Hotspots

INPUTS

- Floorplan db
- Libraries (.ndm)
- SDC file

only for chip to startup (C2S) Training

Goals of Placement by: Modes & Corners

Multicorner-Multimode Concepts

- Designs must often operate under **multiple operating conditions**, often called **corners**, and in **multiple modes**. Such designs are referred to as **multicorner-multimode designs**.
- The following terms are commonly used to describe multicorner-multimode technology:

1. **Corner**
2. **Mode**
3. **Scenario**

Multicorner-Multimode Concepts

Corner:-

- A Corner is defined as a set of libraries characterized for process, voltage and temperature (PVT) variations.

Only for Chip to start

Corners:

- Corners represent variations in process parameters (such as voltage, temperature, and manufacturing tolerances).
- Common corners include:
 - **Fast Process Corner:** Represents the best-case scenario (fastest transistors).
 - **Slow Process Corner:** Represents the worst-case scenario (slowest transistors).
 - **Typical Process Corner:** Represents average conditions.
- Additional corners may arise due to voltage scaling, temperature variations, and other factors.

Mode

Mode :-

A mode is defined by a set of clocks, supply voltages, timing constraints, and libraries. It can also have annotation data, such as SDF or parasitics files.

Multicorner-multimode designs can operate in many modes such as the test mode, functional mode, standby mode and so forth.

Goals of Placement by: Scenarios (Modes & Corners)

- **Scenarios** : Used to perform Timing Analysis and Optimization

Scenarios = Modes + Corners

Recap:

- **Modes** refer to Func mode/ test mode
- **Corners** – rep. diff. Process, Voltage and Temperatures

Only for Chip to startup (C2S) Training